Paper/Subject	B.E/B.Tech.(Paper 1)
Exam Date	29 Jul 2022
Exam Slot	2

Question ID:15477154601
Topic Name: Mathematics-Section A

If $z \neq 0$ be a complex number such that $\left|z-\frac{1}{z}\right|=2$, then the maximum value of $|z|$

Question: is :

A $\sqrt{2}$

B 1

C $\sqrt{2}-1$

D $\sqrt{2}+1$

Answer Given By Candidate:**B**

Question ID:15477154602
Topic Name: Mathematics-Section A

Which of the following matrices can **NOT** be obtained from the matrix $\begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix}$

Question: by a single elementary row operation ?

A $\begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$

B $\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$

C $\begin{bmatrix} -1 & 2 \\ -2 & 7 \end{bmatrix}$

D $\begin{bmatrix} -1 & 2 \\ -1 & 3 \end{bmatrix}$

Answer Given By Candidate:**D**

Question ID:15477154603
Topic Name: Mathematics-Section A

If the system of equations

$$x + y + z = 6$$
$$2x + 5y + \alpha z = \beta$$
$$x + 2y + 3z = 14$$

Question: has infinitely many solutions, then $\alpha + \beta$ is equal to

A 8

B 36

C 44

D 48

Answer Given By Candidate:**D**

Question ID:15477154604
Topic Name: Mathematics-Section A

Let the function $f(x) = \begin{cases} \frac{\log_e(1+5x) - \log_e(1+\alpha x)}{x} & ;if\ x \neq 0 \\ 10 & ;if\ x = 0 \end{cases}$ be continuous at $x = 0$.

Question: Then α is equal to

A 10

B −10

C 5

D −5

Answer Given By Candidate:**Not Attempted**

Question ID:15477154605
Topic Name: Mathematics-Section A

Question: If $[t]$ denotes the greatest integer $\leq t$, then the value of $\int_0^1 \left[2x - \left|3x^2 - 5x + 2\right| + 1\right] dx$ is :

A $\frac{\sqrt{37} + \sqrt{13} - 4}{6}$

B $\frac{\sqrt{37} - \sqrt{13} - 4}{6}$

C $\frac{-\sqrt{37} - \sqrt{13} + 4}{6}$

D $\dfrac{-\sqrt{37}+\sqrt{13}+4}{6}$

Answer Given By Candidate:**Not Attempted**

Question ID:15477154606
Topic Name: Mathematics-Section A

Question: Let $\{a_n\}_{n=0}^{\infty}$ be a sequence such that $a_0 = a_1 = 0$ and $a_{n+2} = 3a_{n+1} - 2a_n + 1, \forall\, n \geq 0.$ Then $a_{25}a_{23} - 2a_{25}a_{22} - 2a_{23}a_{24} + 4a_{22}a_{24}$ is equal to

A 483

B 528

C 575

D 624

Answer Given By Candidate:**Not Attempted**

Question ID:15477154607
Topic Name: Mathematics-Section A

Question: $\sum_{r=1}^{20}\left(r^2+1\right)(r!)$ is equal to

A $22! - 21!$

B $22! - 2\,(21!)$

C $21! - 2\,(20!)$

D $21! - 20!$

Answer Given By Candidate:**Not Attempted**

Question ID:15477154608
Topic Name: Mathematics-Section A

Question: For $I(x) = \int \frac{\sec^2 x - 2022}{\sin^{2022} x}\,dx$, if $I\left(\frac{\pi}{4}\right) = 2^{1011}$, then

A $3^{1010}\, I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$

B

$3^{1010}\, I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$

C $3^{1011}\, I\left(\frac{\pi}{3}\right) - I\left(\frac{\pi}{6}\right) = 0$

D $3^{1011}\, I\left(\frac{\pi}{6}\right) - I\left(\frac{\pi}{3}\right) = 0$

Answer Given By Candidate:**Not Attempted**

Question ID:15477154609
Topic Name: Mathematics-Section A

If the solution curve of the differential equation $\frac{dy}{dx} = \frac{x+y-2}{x-y}$ passes through

Question: the points (2, 1) and (k + 1, 2), k > 0, then

A $2\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(k^2+1\right)$

B $\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(k^2+1\right)$

C $2\tan^{-1}\left(\frac{1}{k+1}\right) = \log_e\left(k^2+2k+2\right)$

D $2\tan^{-1}\left(\frac{1}{k}\right) = \log_e\left(\frac{k^2+1}{k^2}\right)$

Answer Given By Candidate:**D**

Question ID:154771546010
Topic Name: Mathematics-Section A

Let $y = y(x)$ be the solution curve of the differential equation

$\frac{dy}{dx} + \left(\frac{2x^2+11x+13}{x^3+6x^2+11x+6}\right) y = \frac{(x+3)}{x+1}$, $x > -1$, which passes through the point

Question: (0, 1). Then $y(1)$ is equal to :

A $\frac{1}{2}$

B $\frac{3}{2}$

C $\frac{5}{2}$

D $\frac{7}{2}$

Answer Given By Candidate:**Not Attempted**

Question ID:154771546011
Topic Name: Mathematics-Section A

Question: Let m_1, m_2 be the slopes of two adjacent sides of a square of side a such that $a^2 + 11a + 3(m_1^2 + m_2^2) = 220$. If one vertex of the square is $(10(\cos\alpha - \sin\alpha), 10(\sin\alpha + \cos\alpha))$, where $\alpha \in \left(0, \frac{\pi}{2}\right)$ and the equation of one diagonal is $(\cos\alpha - \sin\alpha)x + (\sin\alpha + \cos\alpha)y = 10$, then $72(\sin^4\alpha + \cos^4\alpha) + a^2 - 3a + 13$ is equal to :

A 119

B 128

C 145

D 155

Answer Given By Candidate:**Not Attempted**

Question ID:154771546012
Topic Name: Mathematics-Section A

Question: The number of elements in the set $S = \left\{x \in \mathbb{R} : 2\cos\left(\frac{x^2 + x}{6}\right) = 4^x + 4^{-x}\right\}$ is :

A 1

B 3

C 0

D infinite

Answer Given By Candidate:**Not Attempted**

Question ID:154771546013
Topic Name: Mathematics-Section A

Let A $(\alpha, -2)$, B $(\alpha, 6)$ and C $\left(\frac{\alpha}{4}, -2\right)$ be vertices of a ΔABC. If $\left(5, \frac{\alpha}{4}\right)$ is the circumcentre of ΔABC, then which of the following is **NOT** correct about ΔABC

Question: ?

A area is 24

B perimeter is 25

C circumradius is 5

D inradius is 2

Answer Given By Candidate:**Not Attempted**

Question ID:154771546014
Topic Name: Mathematics-Section A

Let Q be the foot of perpendicular drawn from the point P (1, 2, 3) to the plane $x+2y+z=14$. If R is a point on the plane such that $\angle PRQ=60^\circ$, then the area of ΔPQR is equal to :

Question:

A $\frac{\sqrt{3}}{2}$

B $\sqrt{3}$

C $2\sqrt{3}$

D 3

Answer Given By Candidate:**B**

Question ID:154771546015
Topic Name: Mathematics-Section A

If (2, 3, 9), (5, 2, 1), (1, λ, 8) and (λ, 2, 3) are coplanar, then the product of all possible values of λ is :

Question:

A $\frac{21}{2}$

B $\frac{59}{8}$

C $\frac{57}{8}$

D $\frac{95}{8}$

Answer Given By Candidate:**Not Attempted**

Question ID:154771546016
Topic Name: Mathematics-Section A

Question: Bag I contains 3 red, 4 black and 3 white balls and Bag II contains 2 red, 5 black and 2 white balls. One ball is transferred from Bag I to Bag II and then a ball is drawn from Bag II. The ball so drawn is found to be black in colour. Then the probability, that the transferred ball is red, is :

A $\frac{4}{9}$

B $\frac{5}{18}$

C $\frac{1}{6}$

D $\frac{3}{10}$

Answer Given By Candidate:**D**

Question ID:154771546017
Topic Name: Mathematics-Section A

Question: Let $S = \{z = x+iy : |z-1+i| \geq |z|, |z| < 2, |z+i| = |z-1|\}$. Then the set of all values of x, for which $w = 2x + iy \in S$ for some $y \in \mathbb{R}$, is

A $\left(-\sqrt{2}, \frac{1}{2\sqrt{2}}\right]$

B $\left(-\frac{1}{\sqrt{2}}, \frac{1}{4}\right]$

C $\left(-\sqrt{2}, \frac{1}{2}\right]$

D $\left(-\frac{1}{\sqrt{2}}, \frac{1}{2\sqrt{2}}\right]$

Answer Given By Candidate:**Not Attempted**

Question ID:154771546018
Topic Name: Mathematics-Section A

Let $\vec{a},\vec{b},\vec{c}$ be three coplanar concurrent vectors such that angles between any two of them is same. If the product of their magnitudes is 14 and $(\vec{a}\times\vec{b})\cdot(\vec{b}\times\vec{c})+(\vec{b}\times\vec{c})\cdot(\vec{c}\times\vec{a})+(\vec{c}\times\vec{a})\cdot(\vec{a}\times\vec{b})=168$,then $|\vec{a}|+|\vec{b}|+|\vec{c}|$ is equal

Question: to :

A 10

B 14

C 16

D 18

Answer Given By Candidate:**A**

Question ID:154771546019
Topic Name: Mathematics-Section A

Question: The domain of the function $f(x)=\sin^{-1}\left(\frac{x^2-3x+2}{x^2+2x+7}\right)$ is :

A $[1,\infty)$

B $[-1,2]$

C $[-1,\infty)$

D $(-\infty,2]$

Answer Given By Candidate:**A**

Question ID:154771546020
Topic Name: Mathematics-Section A

Question: The statement $(p\Rightarrow q)\vee(p\Rightarrow r)$ is **NOT** equivalent to

A $(p\wedge(\sim r))\Rightarrow q$

B $(\sim q)\Rightarrow((\sim r)\vee p)$

C $p\Rightarrow(q\vee r)$

D $(p\wedge(\sim q))\Rightarrow r$

Answer Given By Candidate:**B**

Question ID:154771546021
Topic Name: Mathematics-Section B

The sum and product of the mean and variance of a binomial distribution are 82·5 and 1350 respectively. Then the number of trials in the binomial distribution is

Question: _____.

Answer Given By Candidate:**Not Attempted**

Question ID:154771546022
Topic Name: Mathematics-Section B

Let α, β $(\alpha > \beta)$ be the roots of the quadratic equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n$, $n \in N$, then $\dfrac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ is equal to _____.

Question:

Answer Given By Candidate:**Not Attempted**

Question ID:154771546023
Topic Name: Mathematics-Section B

Let $X = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ and $A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -1 \end{bmatrix}$. For $k \in N$, if $X' A^k X = 33$, then k is equal to

Question: _____.

Answer Given By Candidate:**Not Attempted**

Question ID:154771546024
Topic Name: Mathematics-Section B

The number of natural numbers lying between 1012 and 23421 that can be formed using the digits 2, 3, 4, 5, 6 (repetition of digits is not allowed) and divisible by 55 is _____.

Question:

Answer Given By Candidate:**Not Attempted**

Question ID:154771546025
Topic Name: Mathematics-Section B

If $\sum_{k=1}^{10} K^2 \left(10_{C_K}\right)^2 = 22000L$, then L is equal to _____.

Question:

Answer Given By Candidate:**Not Attempted**

Question ID:154771546026
Topic Name: Mathematics-Section B

If $[t]$ denotes the greatest integer $\leq t$, then the number of points, at which the function $f(x) = 4|2x+3| + 9\left[x + \frac{1}{2}\right] - 12[x+20]$ is not differentiable in the open interval (–20, 20), is _____.

Question:

Answer Given By Candidate:**Not Attempted**

Question ID:154771546027
Topic Name: Mathematics-Section B

Question: If the tangent to the curve $y = x^3 - x^2 + x$ at the point (a, b) is also tangent to the curve $y = 5x^2 + 2x - 25$ at the point $(2, -1)$, then $|2a + 9b|$ is equal to _____

Answer Given By Candidate:**195**

Question ID:154771546028
Topic Name: Mathematics-Section B

Question: Let AB be a chord of length 12 of the circle $(x-2)^2 + (y+1)^2 = \frac{169}{4}$.

If tangents drawn to the circle at points A and B intersect at the point P, then five times the distance of point P from chord AB is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:154771546029
Topic Name: Mathematics-Section B

Question: Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}+\vec{b}|^2 = |\vec{a}|^2 + 2|\vec{b}|^2$, $\vec{a}\cdot\vec{b} = 3$ and $|\vec{a}\times\vec{b}|^2 = 75$. Then $|\vec{a}|^2$ is equal to _____.

Answer Given By Candidate:**13**

Question ID:154771546030
Topic Name: Mathematics-Section B

Question: Let $S = \left\{(x,y) \in \mathbb{N}\times\mathbb{N} : 9(x-3)^2 + 16(y-4)^2 \le 144\right\}$ and $T = \left\{(x,y) \in \mathbb{R}\times\mathbb{R} : (x-7)^2 + (y-4)^2 \le 36\right\}$. Then $n(S \cap T)$ is equal to _____

Answer Given By Candidate:**Not Attempted**

Question ID:1269431
Topic Name: Physics-Section A

Question: Two identical metallic spheres A and B when placed at certain distance in air repel each other with a force of F. Another identical uncharged sphere C is first placed in contact with A and then in contact with B and finally placed at midpoint between spheres A and B. The force experienced by sphere C will be:

A 3F/2

B 3F/4

C F

D 2F

Answer Given By Candidate:**D**

Question ID:1269432
Topic Name: Physics-Section A

Match List I with List II.

List I	List II
A. Torque	I. Nms^{-1}
B. Stress	II. $J\,kg^{-1}$
C. Latent Heat	III. Nm
D. Power	IV. Nm^{-2}

Question: Choose the correct answer from the options given below:

A A-III, B-II, C-I, D-IV

B A-III, B-IV, C-II, D-I

C A-IV, B-I, C-III, D-II

D A-II, B-III, C-I, D-IV

Answer Given By Candidate:**A**

Question ID:1269433
Topic Name: Physics-Section A

Question: Two identical thin metal plates has charge q_1 and q_2 respectively such that $q_1 > q_2$. The plates were brought close to each other to form a parallel plate capacitor of capacitance C. The potential difference between them is :

A $\frac{(q_1+q_2)}{C}$

B $\frac{(q_1-q_2)}{C}$

C $\frac{(q_1-q_2)}{2C}$

D $\frac{2(q_1 - q_2)}{C}$

Answer Given By Candidate:**B**

Question ID:1269434
Topic Name: Physics-Section A

Question: Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: Alloys such as constantan and manganin are used in making standard resistance coils.
Reason R: Constantan and manganin have very small value of temperature coefficient of resistance.

In the light of the above statements, choose the ***correct*** answer from the options given below.

A Both **A** and **R** are true and **R** is the correct explanation of **A.**

B Both **A** and **R** are true but **R** is NOT the correct explanation of **A.**

C **A** is true but **R** is false.

D **A** is false but **R** is true.

Answer Given By Candidate:**Not Attempted**

Question ID:1269435
Topic Name: Physics-Section A

Question: A 1 m long wire is broken into two unequal parts X and Y. The X part of the wire is streched into another wire W. Length of W is twice the length of X and the resistance of W is twice that of Y. Find the ratio of length of X and Y.

A 1:4

B 1:2

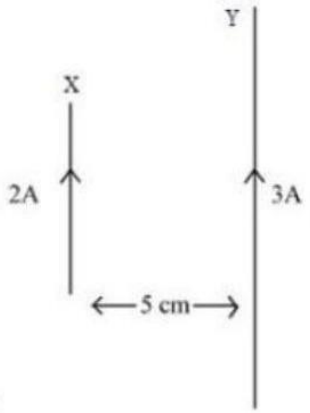
C 4:1

D 2:1

Answer Given By Candidate:**B**

Question ID:1269436
Topic Name: Physics-Section A

A wire X of length 50 cm carrying a current of 2 A is placed parallel to a long wire Y of length 5 m. The wire Y carries a current of 3 A. The distance between two wires is 5 cm and currents flow in the same direction. The force acting on the wire Y is



Question:

A 1.2×10^{-5} N directed towards wire X.

B 1.2×10^{-4} N directed away from wire X.

C 1.2×10^{-4} N directed towards wire X.

D 2.4×10^{-5} N directed towards wire X.

Answer Given By Candidate:**Not Attempted**

Question ID:1269437
Topic Name: Physics-Section A

Question: A juggler throws balls vertically upwards with same initial velocity in air. When the first ball reaches its highest position, he throws the next ball. Assuming the juggler throws n balls per second, the maximum height the balls can reach is

A $g/2n$

B g/n

C $2gn$

D $g/2n^2$

Answer Given By Candidate:**A**

Question ID:1269438
Topic Name: Physics-Section A

Question: A circuit element X when connected to an a.c. supply of peak voltage 100 V gives a peak current of 5 A which is in phase with the voltage. A second element Y when connected to the same a.c. supply also gives the same value of peak current which lags behind the voltage by $\frac{\pi}{2}$. If X and Y are connected in series to the same supply, what will be the rms value of the current in ampere?

A $\frac{10}{\sqrt{2}}$

B $\frac{5}{\sqrt{2}}$

C $5\sqrt{2}$

D $\frac{5}{2}$

Answer Given By Candidate:**Not Attempted**

Question ID:1269439
Topic Name: Physics-Section A

Question: An unpolarised light beam of intensity $2I_0$ is passed through a polaroid P and then through another polaroid Q which is oriented in such a way that its passing axis makes an angle of 30° relative to that of P. The intensity of the emergent light is

A $\frac{I_0}{4}$

B $\frac{I_0}{2}$

C $\frac{3I_0}{4}$

D $\frac{3I_0}{2}$

Answer Given By Candidate:**D**

Question ID:1269440
Topic Name: Physics-Section A

Question: An α particle and a proton are accelerated from rest through the same potential difference. The ratio of linear momenta acquired by above two particals will be:

A $\sqrt{2}:1$

B $2\sqrt{2}:1$

C $4\sqrt{2}:1$

D 8:1

Answer Given By Candidate:**A**

Question ID:1269441
Topic Name: Physics-Section A

Read the following statements:
(A) Volume of the nucleus is directly proportional to the mass number.
(B) Volume of the nucleus is independent of mass number.
(C) Density of the nucleus is directly proportional to the mass number.
(D) Density of the nucleus is directly proportional to the cube root of the mass number.
(E) Density of the nucleus is independent of the mass number.

Question: Choose the correct option from the following options.

A (A) and (D) only.

B (A) and (E) only.

C (B) and (E) only.

D (A) and (C) only

Answer Given By Candidate:**B**

Question ID:1269442
Topic Name: Physics-Section A

An object of mass 1 kg is taken to a height from the surface of earth which is equal to three times the radius of earth. The gain in potential energy of the object will be

Question: [If, $g=10ms^{-2}$ and radius of earth = 6400 km]

A 48 MJ

B 24 MJ

C 36 MJ

D 12 MJ

Answer Given By Candidate:**Not Attempted**

Question ID:1269443
Topic Name: Physics-Section A

A ball is released from a height h. If t_1 and t_2 be the time required to complete first half and second half of the distance respectively. Then, choose the correct relation

Question: between t_1 and t_2.

A $t_1 = \left(\sqrt{2}\right) t_2$

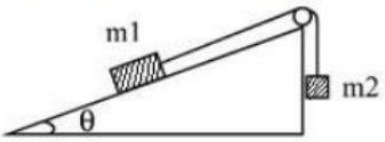
B $t_1 = \left(\sqrt{2} - 1\right) t_2$

C $t_2 = \left(\sqrt{2} + 1\right) t_1$

D $t_2 = \left(\sqrt{2} - 1\right) t_1$

Answer Given By Candidate:**B**

Question ID:1269444
Topic Name: Physics-Section A
Question: Two bodies of masses $m_1 = 5$ kg and $m_2 = 3$ kg are connected by a light string going over a smooth light pulley on a smooth inclined plane as shown in the figure. The system is at rest. The force exerted by the inclined plane on the body of mass m_1 will be :[Take $g = 10\ ms^{-2}$]



A 30 N

B 40 N

C 50 N

D 60 N

Answer Given By Candidate:**A**

Question ID:1269445
Topic Name: Physics-Section A
Question: If momentum of a body is increased by 20%, then its kinetic energy increases by

A 36%

B 40%

C 44%

D 48%

Answer Given By Candidate:**B**

Question ID:1269446
Topic Name: Physics-Section A

Question: The torque of a force $5\hat{i}+3\hat{j}-7\hat{k}$ about the origin is τ. If the force acts on a particle whose position vector is $2\hat{i}+2\hat{j}+\hat{k}$, then the value of τ will be

A $11\hat{i}+19\hat{j}-4\hat{k}$

B $-11\hat{i}+9\hat{j}-16\hat{k}$

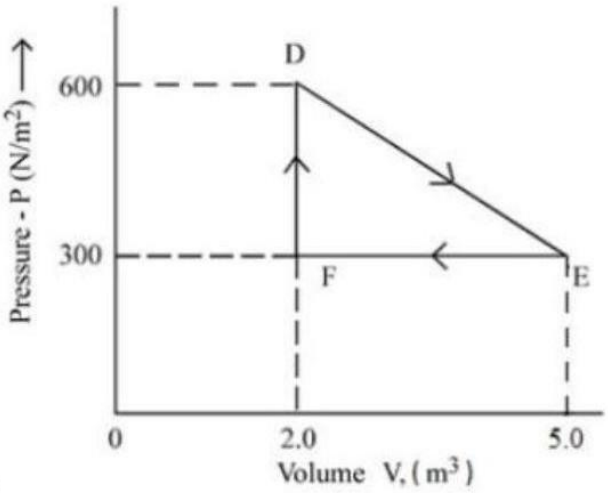
C $-17\hat{i}+19\hat{j}-4\hat{k}$

D $17\hat{i}+9\hat{j}+16\hat{k}$

Answer Given By Candidate:**D**

Question ID:1269447
Topic Name: Physics-Section A

Question: A thermodynamic system is taken from an original state D to an intermediate state E by the linear process shown in the figure. Its volume is then reduced to the original volume from E to F by an isobaric process. The total work done by the gas from D to E to F will be



A −450 J

B 450 J

C 900 J

D 1350 J

Answer Given By Candidate:**B**

Question ID:1269448
Topic Name: Physics-Section A

Question: The vertical component of the earth's magnetic field is 6×10^{-5} T at any place where the angle of dip is 37°. The earth's resultant magnetic field at that place will be (Given $\tan 37° = \frac{3}{4}$)

A 8×10^{-5} T

B 6×10^{-5} T

C 5×10^{-4} T

D 1×10^{-4} T

Answer Given By Candidate:**Not Attempted**

Question ID:1269449
Topic Name: Physics-Section A

Question: The root mean square speed of smoke particles of mass 5×10^{-17} kg in their Brownian motion in air at NTP is approximately. [Given $k = 1.38 \times 10^{-23}$ JK^{-1}]

A 60 mm s^{-1}

B 12 mm s^{-1}

C 15 mm s^{-1}

D 36 mm s^{-1}

Answer Given By Candidate:**Not Attempted**

Question ID:1269450
Topic Name: Physics-Section A

Question: Light enters from air into a given medium at an angle of 45° with interface of the air-medium surface. After refraction, the light ray is deviated through an angle of 15° from its original direction. The refractive index of the medium is :

A 1.732

B 1.333

C 1.414

D 2.732

Answer Given By Candidate:**C**

Question ID:1269451
Topic Name: Physics-Section B

Question: A tube of length 50 cm is filled completely with an incompressible liquid of mass 250 g and closed at both ends. The tube is then rotated in horizontal plane about one of its ends with a uniform angular velocity $x\sqrt{F}$ rad s^{-1}. If F be the force exerted by the liquid at the other end then the value of x will be____.

Answer Given By Candidate:**Not Attempted**

Question ID:1269452
Topic Name: Physics-Section B

Question: Nearly 10% of the power of a 110 W light bulb is converted to visible radiation. The change in average intensities of visible radiation, at a distance of 1 m from the bulb to a distance of 5 m is $a \times 10^{-2}$ W/m^2. The value of 'a' will be _____.

Answer Given By Candidate:**Not Attempted**

Question ID:1269453
Topic Name: Physics-Section B

Question: A metal wire of length 0.5 m and cross-sectional area 10^{-4} m^2 has breaking stress 5×10^8 Nm^{-2}. A block of 10 kg is attached at one end of the string and is rotating in a horizontal circle. The maximum linear velocity of block will be ____ ms^{-1}.

Answer Given By Candidate:**Not Attempted**

Question ID:1269454
Topic Name: Physics-Section B

Question: The velocity of a small ball of mass 0.3 g and density 8 g/cc when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is 1.3 g/cc, then the value of viscous force acting on the ball will be $x \times 10^{-4}$ N, The value of x is _____. [use g = 10m/s^2]

Answer Given By Candidate:**Not Attempted**

Question ID:1269455
Topic Name: Physics-Section B

Question: A modulating signal $2\sin(6.28 \times 10^6)t$ is added to the carrier signal $4\sin(12.56 \times 10^9)t$ for amplitude modulation. The combined signal is passed through a non-linear square law device. The output is then passed through a band pass filter. The bandwidth of the output signal of band pass filter will be __ MHz.

Answer Given By Candidate:**Not Attempted**

Question ID:1269456
Topic Name: Physics-Section B

The speed of a transverse wave passing through a string of length 50 cm and mass 10 g is 60 ms^{-1}. The area of cross-section of the wire is 2.0 mm^2 and its Young's modulus is 1.2×10^{11} Nm^{-2}. The extension of the wire over its natural length due to its tension will be $x \times 10^{-5}$ m. The value of x is